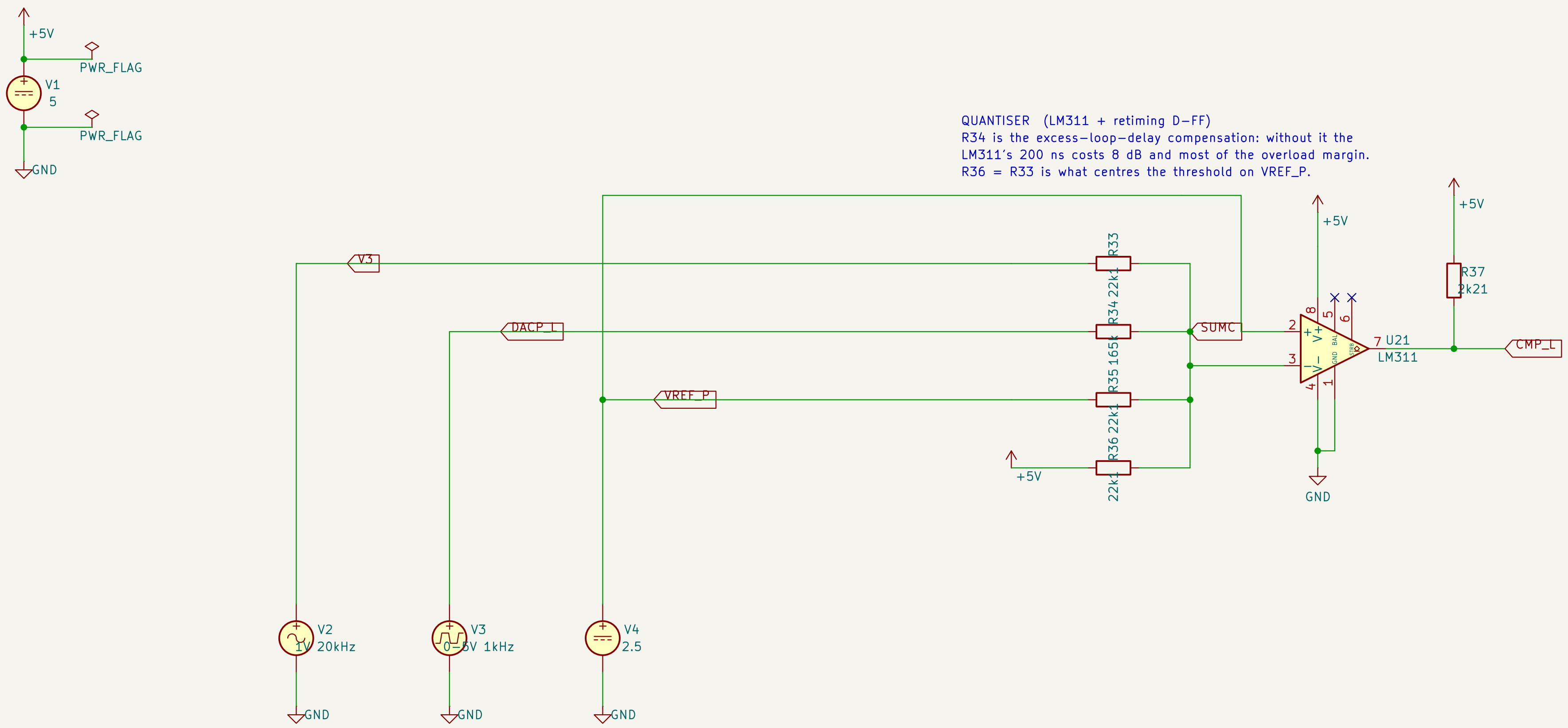
E Quantiser (LM311 on a single +5 V, threshold on VREF_P, and the ELD shift)

QUANTISER (LM311 + retiming D-FF)
R34 is the excess-loop-delay compensation: without it the
LM311's 200 ns costs 8 dB and most of the overload margin.
R36 = R33 is what centres the threshold on VREF_P.



Rb = Rs to +5 V is what centres the summing node on VREF_P:
with the DAC at its mean the comparator switches at V3 = 0,
so the quantiser's threshold costs no offset at all.
Both LM311 inputs then sit at +2.5 V, comfortably inside its
single-supply common-mode range of 0.5 V to V+ − 1.5 V.
Rk0 = 165k is the excess-loop-delay compensation. It moves
the threshold by +/-0.335 V of V3 as the DAC switches, which
is |k0| = 0.335 / S3 = 0.335 / 1.47 = 0.228 -- the design's
-0.225 as an E96 pair. Without it the LM311's 200 ns costs
8 dB and most of the overload margin.

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Title: E — LM311 quantiser on a single supply

Size: A3   Date: Rev:
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